

1. 线性系统

相容性: 有解

等价: 有相同解集

2. 矩阵

增广矩阵

3. 线性系统的图象

① 行图象: n 维空间的超平面

有一个非空交点

consistent $\Leftrightarrow$ 无非空相交区域

② 列图象: 向量的线性组合

4. 矩阵的运算

加减法、数乘

乘法: $(AX)_{(i)} = \sum_{j=1}^n a_{ij} x_j$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$A = [a_{ij}]_{m \times n}$$

$$B = [b_{jk}]_{n \times l}$$

$$AB = \left[\sum_{j=1}^n a_{ij} b_{jk} \right]_{m \times l}$$

线性代数第二周习题课提纲

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1. 回顾上周正课内容

线性系统

- 1) 线性系统的定义: 相容 consistent 有解, 不相容 inconsistent 无解。
- 2) 线性系统的两种解法: 高斯消元法和 Cramer's rule
- 3) 等价线性系统: 有相同的 solution set 的线性系统。(线性系统的 3 种基本操作 改变顺序, 乘以非零数, 乘以非零数然就加到另一个方程里)

矩阵

- 4) 定义系数矩阵 A 和 b 以及增广矩阵 [A|b]
- 5) 矩阵的 3 种基本行操作。通过 3 种基本操作我们可以把矩阵变成行阶梯型 echelon form 以实现求解。

线性系统的图像

- 1) 行图像: 每一行代表一个线/面/超平面。
Consistent $\Leftrightarrow$ 没有非空的相交区域。
- 2) 列图像: 线性系统代表列向量的线性组合。
Consistent $\Leftrightarrow$ 向量 b 位于列向量 span 的区域内。

矩阵的运算

- 1) 矩阵的加减法与数乘的定义。矩阵的简写 $A = [a_{ij}]_{m \times n}$, size $m \times n$ 。
- 2) 用线性方程组定义矩阵与向量的乘法。
- 3) 定义矩阵与矩阵的一般乘法规则 (把后一个矩阵看做是多个列向量构成的行向量)。
- 4) 矩阵乘法的不同看法:
AB 的每一个元素可看做前一个矩阵 A 的行与后一个矩阵 B 的列的内积。
AB 的每一列都可以看做 A 与 B 的对应列的积。

Vector space (Sheldon Axler)

1.19 **Definition** vector space

A vector space is a set V along with an addition on V and a scalar multiplication on V such that the following properties hold:

- commutativity**
 $u + v = v + u$ for all $u, v \in V$;
- associativity**
 $(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$ for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$;
- additive identity**
there exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$;
- additive inverse**
for every $v \in V$, there exists $w \in V$ such that $v + w = 0$;
- multiplicative identity**
 $1v = v$ for all $v \in V$;
- distributive properties**
 $a(u + v) = au + av$ and $(a + b)v = av + bv$ for all $a, b \in \mathbb{F}$ and all $u, v \in V$.

Span (Sheldon Axler)

2.5 **Definition** span

The set of all linear combinations of a list of vectors $v_1, \dots, v_m$ in V is called the *span* of $v_1, \dots, v_m$, denoted $\text{span}(v_1, \dots, v_m)$. In other words,

$$\text{span}(v_1, \dots, v_m) = \{a_1 v_1 + \dots + a_m v_m : a_1, \dots, a_m \in \mathbb{F}\}.$$

The span of the empty list $()$ is defined to be $\{0\}$.

Vector space (Gilbert Strang)

geometry is hard to visualize. Within all vector spaces, two operations are possible:

We can add any two vectors, and we can multiply all vectors by scalars. In other words, we can take linear combinations.

高斯消元练习题

30. Use elimination to solve

$$\begin{array}{rcl} u + v + w & = & 6 \\ u + 2v + 2w & = & 11 \\ 2u + 3v - 4w & = & 3 \end{array} \quad \text{and} \quad \begin{array}{rcl} u + v + w & = & 7 \\ u + 2v + 2w & = & 10 \\ 2u + 3v - 4w & = & 3. \end{array}$$

31. For which three numbers a will elimination fail to give three pivots?

$$ax + 2y + 3z = b_1$$

$$ax + ay + 4z = b_2$$

$$ax + ay + az = b_3.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & -7 & -14 \end{array} \right] \quad \begin{array}{l} u=1 \\ v=3 \\ w=2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -7 & -14 \end{array} \right] \quad \begin{array}{l} u=4 \\ v=1 \\ w=2 \end{array}$$

$$\left[\begin{array}{ccc|c} a & 2 & 3 & b_1 \\ 0 & a-2 & 1 & b_2-b_1 \\ 0 & 0 & a-4 & b_3-b_2 \end{array} \right] \quad a=0 \text{ or } 2 \text{ or } 4$$

$$x \begin{bmatrix} a \\ a \\ a \end{bmatrix} + y \begin{bmatrix} 2 \\ a \\ a \end{bmatrix} + z \begin{bmatrix} 3 \\ 4 \\ a \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{v}_1 = \alpha \vec{v}_2 + \beta \vec{v}_3 \text{ (共面)}$$

$$\alpha \begin{bmatrix} 2 \\ a \\ a \end{bmatrix} + \beta \begin{bmatrix} 3 \\ 4 \\ a \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix}$$

29. (Very optional) Normally the multiplication of two complex numbers

$$(a+ib)(c+id) = (ac-bd) + i(bc+ad)$$

involves the four separate multiplications ac, bd, bc, ad . Ignoring i , can you compute $ac-bd$ and $bc+ad$ with only three multiplications? (You may do additions, such as forming $a+b$ before multiplying, without any penalty.)

$$\begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} c & d \end{bmatrix} = \begin{bmatrix} ac & ad \\ bc & bd \end{bmatrix} \quad \begin{array}{l} ac-bd \quad bc+ad \\ (a-b)(c+d) = ac+ad-bc-bd \end{array}$$

$$\begin{bmatrix} a+b \\ c+d \end{bmatrix} \begin{bmatrix} a+b & c+d \end{bmatrix} \quad \begin{array}{l} (a+b)(c-d) = ac-ad+bc-bd \\ (a+c)(b+d) = ab+ad+bc+cd \end{array}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ac-bd \\ bc+ad \end{bmatrix}$$

$$a_{21}=0 \quad a_{22}=a_{23}=1 \quad y=bc \quad z=ad \quad (\cancel{y} \quad bc+ad)$$

$$a_{11}x + a_{12}bc + a_{13}ad = ac-bd \quad (\cancel{a_{11}}=1)$$

$$\therefore x = ac-bd - a_{12}bc - a_{13}ad$$

$$= a(c - a_{13}d) - b(a_{12}c + d)$$

$$\therefore \begin{cases} a_{13}=1 & a_{12}=-1 & x=(a+b)(c-d) \\ \text{or} \\ a_{13}=-1 & a_{12}=1 & x=(a-b)(c+d) \end{cases}$$

$$AB = A(b_1 \dots b_n) = (Ab_1 \dots Ab_n)$$

$$(Ax)_{ij} = \sum_{j=1}^n a_{ij} x_j$$

$$B(Ax) = \sum_{i=1}^m b_{ki} \left(\sum_{j=1}^n a_{ij} x_j \right)$$

$$= \sum_{i=1}^m \sum_{j=1}^n b_{ki} a_{ij} x_j$$

$$\downarrow$$

$$Mx = \sum_{j=1}^n m_{kj} x_j \quad m_{kj} = \sum_{i=1}^m b_{ki} a_{ij}$$

$$AB = A(b_1 \dots b_n) = (Ab_1 \dots Ab_n)$$

$$(Ax)_{ij} = \sum_{j=1}^n a_{ij} x_j$$

$$\downarrow$$

$$B(Ax) = \sum_{i=1}^m b_{ki} \left(\sum_{j=1}^n a_{ij} x_j \right)$$

$$\downarrow$$

$$Mx = \sum_{j=1}^n m_{kj} x_j \quad m_{kj} = \sum_{i=1}^m b_{ki} a_{ij}$$

矩阵的不同视角:

① 以 i, j 为变量的函数

② 列向量的行 / 行向量的列

2. 证明: 若齐次线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases}$$

中方程的个数小于未知元的个数, 也即 $m < n$, 则方程含有无穷多个解。

1. 证明: 齐次线性方程组

$$\begin{cases} a_{11}x + a_{12}y = 0 \\ a_{21}x + a_{22}y = 0 \end{cases}$$

视为向量

有非零解 (即 x 和 y 不全为 0) 的充分必要条件是 $a_{11}a_{22} - a_{12}a_{21} = 0$.

1. 矩阵运算的性质

① 矩阵乘法: $(AB)C = A(BC)$ $AB \neq BA$

② 矩阵加法: $(A+B)+C = A+(B+C)$

$$A+B = B+A$$

$$(A+B)C = AC + BC$$

③ 矩阵数乘: $(\alpha\beta)A = \alpha(\beta A)$

$$(\alpha A)B = A(\alpha B)$$

$$(\alpha + \beta)A = \alpha A + \beta A$$

2. LU分解

① 初等矩阵

Type I $\begin{bmatrix} 1 & & \\ & 1 & \\ & & \ddots \end{bmatrix}$, Type II $\begin{bmatrix} 1 & & \\ & \alpha & \\ & & \ddots \end{bmatrix}$, Type III $\begin{bmatrix} 1 & & \\ & 1 & \\ & & \ddots \end{bmatrix}$

② LU分解

① 用 Type III $GFEA = U$

② $A = E^{-1}F^{-1}G^{-1}U$

③ LDU分解: $U = DU'$ $A = LDU'$

3. 分块矩阵

① 特例: $B = [b_1, b_2, \dots, b_n]$

$$AB = [Ab_1, Ab_2, \dots, Ab_n]$$

$$A = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, AB = \begin{bmatrix} a_1 B \\ \vdots \\ a_n B \end{bmatrix}$$

② 一般规则:

(1) 前一个矩阵的块的列数与后一个矩阵的块的行数相等

(2) 前一个块矩阵的列数与后一个矩阵的块的行数相等

线性代数第二周习题课提纲

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1. 回顾上周正课内容

矩阵乘积的一些性质

- 1) 矩阵乘法：结合律，一般不满足交换率。
 - 2) 矩阵加法：交换律，结合率，（加法和乘法的）分配率。
 - 3) 矩阵数乘：交换律，结合律，（数的加法和数乘的）分配率。
- 总结：除了矩阵乘法不满足交换律，矩阵的运算法则和数的运算法则一样。

矩阵的 LU 分解

- 1) 对应矩阵的初等变换的 3 种初等矩阵 (elementary matrix)
Type I: 交换单位矩阵的两行。（左乘操作行，右乘操作列。）
Type II: 将单位矩阵的特定行乘以一个数。（左乘操作行，右乘操作列。）
Type III: 将单位矩阵的一行乘以一个数加到另一行上。（...）
- 2) 方形矩阵的 LU 分解的步骤：
① 用 Type III 初等行变换逐步将矩阵变成上三角阵（即 U 矩阵）。
② 求出各个初等行变换的逆矩阵乘，将逆矩阵乘在一起（即 L 矩阵）。
- 3) LDU 分解：
LU 分解不唯一，但 LDU 分解唯一。LDU 分解即将 LU 分解后的 U 分解成对角矩阵和对角项为 1 的上三角矩阵。

分块矩阵 (Partitioned Matrix)

- 1) 特例：
① AB 的每一列都可以看做 A 与 B 的对应列的乘积。
② AB 的每一行都可以看做 A 的对应行与 B 的乘积。
- 2) 一般分法：
分块矩阵 A 与 B 的对应块的列与行数需要相等；分块之后的前一个块矩阵的列数与行数要相等。

1. 证明：齐次线性方程组

$$\begin{cases} a_{11}x + a_{12}y = 0 \\ a_{21}x + a_{22}y = 0 \end{cases}$$

有非零解（即 x 和 y 不全为 0）的充分必要条件是 $a_{11}a_{22} - a_{12}a_{21} = 0$ 。

$$\textcircled{1} a_{11} = a_{21} = 0 \quad \begin{cases} a_{12}y = 0 \\ a_{22}y = 0 \end{cases} \text{ 有非零解} \Rightarrow a_{11}a_{22} - a_{12}a_{21} = 0$$

$$\textcircled{2} a_{11} \neq 0, a_{21} = 0 \quad \begin{cases} a_{11}x + a_{12}y = 0 \\ a_{22}y = 0 \end{cases} \text{ 有非零解} \Rightarrow a_{22} = 0 \Rightarrow a_{11}a_{22} - a_{12}a_{21} = 0$$

$$a_{11}a_{22} - a_{12}a_{21} = 0 \Rightarrow a_{22} = 0 \Rightarrow \text{有非零解}$$

$$\textcircled{3} a_{11} \neq 0, a_{21} \neq 0 \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{22} - \frac{a_{21}a_{12}}{a_{11}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ 有非零解}$$
$$\Downarrow$$
$$a_{22} - \frac{a_{12}a_{21}}{a_{11}} = 0 \Rightarrow a_{11}a_{22} - a_{12}a_{21} = 0$$

2. 证明: 若齐次线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases}$$

中方程的个数小于未知元的个数，也即 $m < n$ ，则方程含有无穷多个解。

$$b_1x_1 + \dots + b_nx_n = 0$$

$$b_{22}x_2 + \dots + b_{n2}x_n = 0$$

$$b_{rr}x_r + \dots + b_{rn}x_n = 0$$

$$0 = 0$$

$$Q=0$$

$$\begin{bmatrix} b_{11} + \dots + b_{1r} \\ \vdots \\ b_{rr} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_r \end{bmatrix} =$$

$$\begin{bmatrix} -b_{rr+1}x_{r+1} - b_{rn}x_n \\ \vdots \\ -b_{rr+1}x_{r+1} - b_{rn}x_n \end{bmatrix}$$

剩余项
(移动到右侧)

$(n-r)$ 个未知数

5. 求所有 2×2 矩阵 A , 使得

$$A^2 = O$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & d^2+bc \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$a^2 = d^2 \quad a = d \text{ or } a = -d$$

① $a=d=0 \Rightarrow bc=0 \Rightarrow A = \begin{bmatrix} & b \\ c & \end{bmatrix}$ $bc=0$

② $a=d \neq 0 \Rightarrow bc < 0$, $b \neq 0$ 且 $c \neq 0 \Rightarrow$ 不存在 A^2 (右上左下不能同时为0)

③ $a = -d \neq 0 \Rightarrow bc < 0, b(a+d) = c(a+d) = 0, bc = -a^2$

↓

$$A = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}, \quad bc = -a^2 \Leftarrow$$

矩阵的逆:

$$A, A^{-1}, A^{-1}A = AA^{-1} = I$$

唯一性: $A^{-1} \quad (AB)^{-1} = B^{-1}A^{-1} \quad B^{-1}A^{-1}AB = I$

① 定理: $Ax=0$ 有非零解 $\Rightarrow A$ 不可逆

A 可逆: $Ax=0 \quad A^{-1}(Ax) = A^{-1}0 \Rightarrow x = A^{-1}0 = 0$

② 定理: 2×2 矩阵可逆 $\Leftrightarrow ad-bc \neq 0$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow ad-bc=0 \Leftrightarrow Ax=0 \text{ 有非零解} \Rightarrow A \text{ 不可逆}$$

$$\Leftarrow ad-bc \neq 0$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

③ 方阵的逆矩阵的计算 (Gauss-Jordan)

$$AA^{-1} = I_n = [Ax_1, Ax_2, \dots, Ax_n]$$

④ 性质: A 可逆 $\Leftrightarrow A$ 有完整一组 pivots

⑤ 非奇异 (non-singular) $\forall b, Ax=b$ 只有一个解 (非奇异), 则 A 为非奇异的

⑥ A 可逆 $\Leftrightarrow A$ 非奇异 $\Rightarrow Ax=b \quad A^{-1}(Ax) = A^{-1}b$

$$\Rightarrow x = A^{-1}b \Rightarrow A \text{ 非奇异}$$

转置矩阵: $A = (a_{ij})_{m \times n} \quad A^T = (b_{ij})_{n \times m} \quad b_{ij} = a_{ji}$

$$(A^T)^T = A \quad (\alpha A)^T = \alpha A^T \quad (A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T \quad (A^T)^{-1} = (A^{-1})^T$$

对称矩阵: $A = A^T$ (方阵)

$$(A^{-1})^T = A^{-1} \quad (RR^T)^T = (R^T)^T R^T = RR^T$$

定理: $A = A^T$ $A = LDU$ $L^T = U$

$$A = A^T = (LDU)^T = U^T D L^T$$

1. 证明: 两个 n 阶 (n 行 n 列) 下三角矩阵的乘积仍为下三角矩阵. 进一步, 若下三角矩阵的主对角元素都为 1, 则其乘积的主对角线元素也为 1.

$$A = (a_{ij})_{n \times n}, \text{ if } i < j, a_{ij} = 0$$

$$B = (b_{jk})_{n \times n}, \text{ if } j < k, b_{jk} = 0$$

$$(AB)_{ik} = \sum_{j=1}^n a_{ij} b_{jk} \text{ (证明在 } k > i \text{ 时, } (AB)_{ik} = 0 \text{)}$$

$$\text{令 } k > i, \text{ ①若 } j > i, a_{ij} = 0, \checkmark$$

$$\text{②若 } j \leq i, j \leq i < k, b_{jk} = 0, \checkmark$$

$$k = i, \text{ ① } j > i, a_{ij} = 0, \checkmark$$

$$\text{② } j < i, j < i = k, b_{jk} = 0, \checkmark$$

$$\text{③ } j = i, j = i = k, a_{ii} b_{ii} \neq 0, \text{ 其余均为 } 0, \checkmark$$

$$\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}^n = \left(\begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)^n = \begin{bmatrix} 2 & 3 \end{bmatrix} 8^{n-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

5. 如果 n 阶方阵 A 可以跟所有的 n 阶方阵交换, 也即 $AB = BA$, 对任意的 n 阶方阵 B 成立, 则 $A = \lambda I_n$, 其中 λ 是一个实数, I_n 是 n 阶恒等 (单位) 矩阵.

设 $B_{ii} = 1$, 其余为 0 $BA = AB \Rightarrow$

$$\begin{bmatrix} a_{11} & 0 & a_{1n} \\ 0 & a_{11} & 0 \\ 0 & a_{1n} & 0 \end{bmatrix}$$

当 $i \neq j$, $a_{ij} = 0$ $A = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{11} & 0 \\ 0 & 0 & a_{11} \end{bmatrix}$

$B_{ii} = 1 \Rightarrow a_{ii} = a_{jj} \Rightarrow A = \lambda I_n$

列空间: $A = [a_1, a_2, \dots, a_n] \quad \{x_1 a_1 + x_2 a_2 + \dots + x_n a_n \mid x_i \in \mathbb{R}\}$

$$Ax=b \text{ 可解} \Leftrightarrow b \in C(A)$$

nullspace: $\{x \mid Ax=0\}$

若 x_p 是 $Ax=b$ 的一个解, $x = x_p + x_N$ ($x_N \in N(A)$)

最简行阶梯形: 

$$X = V_1 + aV_2 + bV_3$$

矩阵的秩: 线性无关的行向量的数目 $\leq$ # of pivots
($n-r$) free variables

线性无关: $V_1, V_2, \dots, V_n \in V$

$$a_1 V_1 + a_2 V_2 + \dots + a_n V_n = 0 \Leftrightarrow a_1 = \dots = a_n = 0$$

$$\Leftrightarrow \exists a_i \neq 0$$

$$-a_j V_j = a_1 V_1 + \dots + a_{i-1} V_{i-1} + a_{i+1} V_{i+1} + \dots$$

$V_i = -\frac{1}{a_i} (\text{——})$ 线性相关: 其中某个向量可以表示为其他向量的线性组合

定理: if A 的 column vector linearly independent

$$\Leftrightarrow N(A) = \{0\}, \quad x_1 a_1 + x_2 a_2 + \dots + x_n a_n = 0$$

1. 设 V 为一实向量空间，也即它是一集合并定义了向量加法和数乘运算，它们满足向量空间的 8 条公理，请从这 8 条公理出发推导下面的性质 (这里 $\alpha, \beta \in \mathbb{R}, x, y, z \in V$):

不用减和除

1. $(-1) \cdot x = -x, \forall x \in V$;
2. 若 $x + y = x + z$, 则 $y = z$;
3. $\beta \cdot \vec{0} = \vec{0}, \forall \beta \in \mathbb{R}$;
4. 若 $\alpha \cdot x = \vec{0}$, 则 $\alpha = 0$ 或者 $x = \vec{0}$.

$$1. (-1) \cdot x + x = \vec{0} \Rightarrow (-1) \cdot x = -x$$

$$\Downarrow \\ = (-1+1)x = \underline{0 \cdot x = \vec{0}}$$

证明过的推论: $0 \cdot x + x = (0+1)x = x$

$$x + (-1) \cdot x = \vec{0} \text{ 可证 } (x+y=y+x)$$

$$2. x+y+(-x) = x+z+(-x) \Rightarrow y=z$$

$$3. \beta \cdot \vec{0} + x = x$$

$$\Downarrow \\ = \beta \cdot \vec{0} + (\beta \cdot \frac{1}{\beta})x = \beta(\vec{0} + \frac{1}{\beta}x) = \beta \cdot \frac{1}{\beta}x = x.$$

$$4. \alpha x = \vec{0}$$

$$i. \text{ 令 } \alpha \neq 0, x = (\frac{1}{\alpha} \cdot \alpha)x = \frac{1}{\alpha}(\alpha x) = \vec{0}$$

$$ii. \text{ 令 } \alpha = 0, \checkmark$$

2. 设 n 阶矩阵 A 满足 $A^2 + 2A - 3I_n = 0$.

(a) 证明 $A, A + 2I_n$ 可逆, 并求它们的逆.

(b) 当 $A \neq I_n$ 时, 判断 $A + 3I_n$ 是否可逆, 并说明理由.

$$(a) \frac{1}{3}A(A+2I) = I$$

$$(b) \underbrace{(A+3I)}_{\uparrow} \underbrace{(A-I)}_{\neq 0} = 0$$

不能说明是零矩阵 ($A^2=0, A \neq 0$ is possible)

假设 $A+3I$ invertible,

$$(A+3I)^{-1}(A+3I)(A-I) = A-I = (A+3I)^{-1}0 = 0,$$

与 $A \neq I$ 矛盾.

or 证 $(A+3I)V_i = \vec{0}, i=1,2,\dots,n$ 有非零解

$Ax=b$ 有唯一解 $\Leftrightarrow A$ 可逆

A 不可逆 $\Leftrightarrow Ax=b$ 不止一解

$Ax=0$ 有非零解

3. 设有 n 阶方阵 $A = (a_{ij})$ 满足

$$2|a_{ii}| > \sum_{j=1}^n |a_{ij}|,$$

对任意的 $i = 1, 2, \dots, n$ 成立, 证明 A 为可逆矩阵.

假设 A 不可逆, $\Rightarrow Ax=0$ 有非零解

设 $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, $|x_k| \geq |x_i|$ ($i = 1, 2, \dots, n$)

$$a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kn}x_n = 0$$

$$\Downarrow$$
$$a_{k1}x_1 + \dots + a_{kn}x_n = -a_{kk}x_k$$

$$\left| \sum_{i \neq k} a_{ki}x_i \right| = |a_{kk}x_k|$$

$$\sum_{i \neq k} |a_{ki}| |x_i| \geq |a_{kk}| |x_k| \Rightarrow \sum_{i \neq k} |a_{ki}| \geq |a_{kk}|$$

$$\Downarrow$$
$$\sum |a_{ki}| \geq 2|a_{kk}|, \text{ 矛盾}$$

1. 线性相关 $V_1 \dots V_n$ $C_1 V_1 + C_2 V_2 + \dots + C_n V_n = 0 \Leftrightarrow \exists C_i \neq 0$
无关 $\Leftrightarrow \forall C_i = 0$

2. Span: V , $\forall V \in V$, $V = C_1 V_1 + \dots + C_n V_n$ $V = \text{span}\{V_1, \dots, V_n\}$

3. basis: $V_1, \dots, V_n$ ① $V = \text{span}\{V_1, \dots, V_n\}$

② $V_1, \dots, V_n$ 线性无关

① $\forall V = C_1 V_1 + \dots + C_n V_n = C'_1 V_1 + \dots + C'_n V_n$ $C_i = C'_i$

(相减证明)

② if $\{V_1, \dots, V_n\}$ 与 $\{W_1, \dots, W_n\}$ 是 V 的基, $n=m$

basis 是一组基, basis vector 一般指一个向量

4. dimension basis 向量数

$\mathbb{R}^n$ $\{V_1, \dots, V_n\}$

$\{V_1, \dots, V_n\}$ 是 $\mathbb{R}$ 的基 $\Leftrightarrow [V_1 \ V_2 \ \dots \ V_n]$ invertible

5.

	dim	basis
$C(A)$	r	pivot columns
$N(A)$	$n-r$	free variables 逐个取1其余取0
$C(A^T)$	r	$C(A^T) = C(U)$ pivot rows
$N(A^T)$	$m-r$	LU decomposition L^{-1} 的最底部 $(m-r)$ 行

6. 秩-零化度定理: $\dim C(A) + \dim N(A) = n$

7. $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$

8. 非方阵 $\exists B, BA=I$ (A 存在左逆) $\Leftrightarrow \text{rank}(A)=n$
 $\exists B, AB=I$ (A 存在右逆) $\Leftrightarrow \text{rank}(A)=m$

1. 线性方程组

$$Ax = b$$

有解当且仅当 A 的秩等于增广矩阵 $[A \ b]$ 的秩，也即

$$\text{rank}(A) = \text{rank}([A \ b])$$

这里增广矩阵是指把 b 作为列向量添加到 A 中成为最后一列.

$$Ax=b \text{ 有解} \Leftrightarrow b = x_1 a_1 + \dots + x_n a_n, A = [a_1 \dots a_n] \\ \Leftrightarrow b \in C(A) \Leftrightarrow \dim C(A) = \dim C[A \ b] \\ r(A) = r(A \ b)$$

2. 已知 3×3 矩阵 A 的第一行元素不全为零，矩阵

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 7 \end{bmatrix}$$

满足 $AB = O$ (这里 O 是 3 乘 3 的零矩阵)，求线性方程组 $Ax = 0$ 的通解.

$$r(A) \geq 0 \text{ (第一行不全为0)}$$

$$\dim N(A) = n - r(A) \leq n - 1 = 2$$

$$\therefore v_1, v_2 \text{ 线性无关} \Rightarrow \dim N(A) = 2$$

3. 设 W_1, W_2 为向量空间 V 的两个子空间，问

(a) $W_1 \cap W_2 := \{x \in V | x \in W_1 \text{ 且 } x \in W_2\}$ 是否为 V 的子空间. 阐明理由.

(b) $W_1 + W_2 := \{w_1 + w_2 | w_1 \in W_1, w_2 \in W_2\}$ 是否为子空间. 阐明理由.

验证负向存在子空间

4. 已知非齐次线性方程组

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = -1 \\ 4x_1 + 3x_2 + 5x_3 - x_4 = -1 \\ ax_1 + x_2 + 3x_3 + bx_4 = 1 \end{cases}$$

(a) 证明方程组的系数矩阵的秩为 2.

(b) 求 a, b 的值和方程组的通解.

有 3 个线性无关的解.

(不是只有)

$$X = X_p + \underset{\substack{\uparrow \\ n-r}}{X_n} \quad X_n \in N(A)$$

$$1+n-r \geq 3$$

$$5-r \geq 3$$

$$\begin{cases} r \leq 2 \\ r \geq 2 \end{cases}$$

$$(已知前2个方程线性无关) \Rightarrow r=2$$

$$\begin{bmatrix} 1 & 4 \\ 1 & 3 \\ 1 & 5 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} a \\ 1 \\ 3 \\ b \\ 1 \end{bmatrix}$$

$r=2,$
(V_3 与 V_1, V_2 一定线性相关)

5. 求以下齐次线性方程组的通解:

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 + x_3 = 0 \\ \vdots \\ x_{n-1} + x_n = 0 \\ x_n + x_1 = 0 \end{cases}$$

$$\begin{aligned} n \text{ 偶: } x_n &= -x_1 \Rightarrow \begin{bmatrix} x_1 \\ -x_1 \\ \vdots \\ -x_1 \end{bmatrix} \\ x_n &= (-1)^{n-1} x_1 \\ x_n &= -x_1 \end{aligned}$$

$$n \text{ 奇: 全 } 0$$

线性变换:

$$1. \text{定义 } T: V \rightarrow W \begin{cases} \textcircled{1} T(v+w) = T(v) + T(w) \\ \textcircled{2} T(\alpha v) = \alpha T(v) \end{cases}$$

(vector space)

$$2. \text{性质: } T(\vec{0}) = \vec{0}, T(av+bw) = aT(v) + bT(w)$$

$$3. \text{引理: } \{v_1, v_2\} \text{ 是 } V \text{ 的一个 basis, } T: V \rightarrow W,$$

$$T(v_i) (\forall i=1, \dots, n) \Rightarrow T(v), \forall v \in V$$

线性变换的矩阵表示

$$1. \text{定义: } V: \{v_j\}, W: \{w_j\}$$

$$T: V \rightarrow W \text{ 可被 } A \text{ 表示为 } A = [a_{ij}]_{m \times n},$$

$$T(v_j) = a_{1j}w_j + a_{2j}w_j + \dots + a_{mj}w_j$$

$$2. T([v_1 \dots v_n]) = [w_1 \dots w_n] A$$

$$v = [v_1 \dots v_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad T: \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \rightarrow A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$3. \left. \begin{array}{l} S \text{ 表示 } A \\ T \text{ 表示 } B \end{array} \right\} A \circ B \text{ 被 } ST \text{ 表示}$$

$$4. T: V \rightarrow V \begin{cases} T([v_1 \dots v_n]) = [v_i] A \\ T([w_i]) = [w_i] B \\ [w_i] = [v_i] S \end{cases}$$

$$B = S^{-1}AS$$

$$T([v_i]S) = T([v_i])S = [v_i]AS$$

1. 设

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 3 & 0 \\ 1 & 0 & 2 & 1 \\ 3 & 2 & 8 & 0 \end{bmatrix}$$

(a) 找出列空间 $C(A)$ 的一组基.

(b) 写出一个矩阵 B 使得 $C(A) = N(B)$. (请解释为何你给出的矩阵满足要求)

$$r(A)=3 \quad n(B)=4$$

$$3 = 4 - \text{rank}(B) \Rightarrow \text{rank}(B)=1$$

$$\Rightarrow [b_1 \dots b_4] \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 2 & 0 \end{bmatrix}$$

2. 设 A 为一个秩为 1 的 3 阶矩阵,

$$B = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 4 & k \\ 5 & 5 & 15 \end{bmatrix}$$

如果 $AB = O$, 其中 O 为零矩阵, 求 k 的值.

$$r(A)=1 \quad \dim N(A)=2$$

3. 设 A 为任意矩阵, 证明: $\text{rank}(A^T A) = \text{rank}(A)$.

$$N(A) \subseteq N(A^T A) \quad \underline{\text{显然}}$$

$$N(A) \supseteq N(A^T A):$$

$$A^T A x = 0 \quad x^T A^T A x = 0$$

$$(Ax)^T (Ax) = 0$$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$c_1^2 + \dots + c_n^2 = 0$$

$$\Downarrow$$
$$c_1 = \dots = c_n = 0$$

4. 证明: 对于任意两个大小相同的矩阵 A, B , 均有

$$\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B).$$

再举例说明 $\text{rank}(A+B) < \text{rank}(A) + \text{rank}(B)$ 和 $\text{rank}(A+B) = \text{rank}(A) + \text{rank}(B)$ 均有可能出现.

$$\begin{aligned} r \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} &= r \begin{bmatrix} A & A \\ 0 & B \end{bmatrix} \\ &= r \begin{bmatrix} A & A \\ A & A+B \end{bmatrix} \\ &\geq \text{rank}(A+B) \end{aligned}$$

$$C(A+B) \subseteq \text{span} \left\{ \underbrace{a_1, \dots, a_q}_{\text{basis of } C(A)}, \underbrace{b_1, \dots, b_p}_{\text{basis of } C(B)} \right\}$$

$$\Rightarrow \dim C(A+B) \leq q+p = \dim C(A) + \dim C(B)$$

5. 假设 A, B 矩阵的大小都为 $m \times n$ 并且它们的秩都小于 $\frac{n}{2}$, 证明:
 $Ax=0$ 和 $Bx=0$ 这两个齐次线性方程组一定有非零的公共解.

$$\textcircled{1} r(A) < \frac{n}{2}, \quad r(B) < \frac{n}{2}$$

$$\begin{cases} Ax=0 \\ Bx=0 \end{cases} \quad \begin{bmatrix} A \\ B \end{bmatrix} x = 0$$

消元至少剩 $r(A)+r(B)$ 非 0 行

$$\Rightarrow r \begin{bmatrix} A \\ B \end{bmatrix} \leq r(A) + r(B) < n$$

$$\dim(N \begin{bmatrix} A \\ B \end{bmatrix}) = n - r > 0 \Rightarrow \text{有非零公共解}$$

$$\textcircled{2} \dim N(A) > \frac{n}{2}, \quad \dim N(B) > \frac{n}{2}.$$

$$N(A) = \text{span} \{a_1, \dots, a_q\}, \quad N(B) = \text{span} \{b_1, \dots, b_p\}$$

$$\{a_1, \dots, a_q, b_1, \dots, b_p\} \quad \# > n \quad \text{线性相关}$$

$$\therefore \exists a_q = \alpha_1 a_1 + \dots + \alpha_{q-1} a_{q-1} + \beta_1 b_1 + \dots + \beta_p b_p.$$

$$\text{必有 } \exists \beta_i \neq 0, \quad \underbrace{\beta_1 b_1 + \dots + \beta_p b_p}_{\neq 0} \quad (a_i \text{ 为 } a_q)$$

1. 长度: $\|x\| = \sqrt{x^T x}$

正交: $x^T y = 0$

$x \perp y, \|x - y\|^2 = \|x\|^2 + \|y\|^2$

$$\|x^T y\|^2 \leq \|x\|^2 \|y\|^2$$

命题: $\{v_i\}$ 正交 $\Rightarrow \{v_i\}$ 线性独立

$\{v_i\}$ 正交且 $V = \text{span}\{v_i\}$: 正交基

$V \subseteq U, W \subseteq U: \forall v \in V, w \in W, v \perp w \Rightarrow V \perp W$

$$\begin{cases} C(A) \perp N(A^T) \\ C(A^T) \perp N(A) \end{cases}$$

正交补空间: $V \subseteq U, \{w \mid \forall w \in U, w \perp V\} \equiv V^\perp$

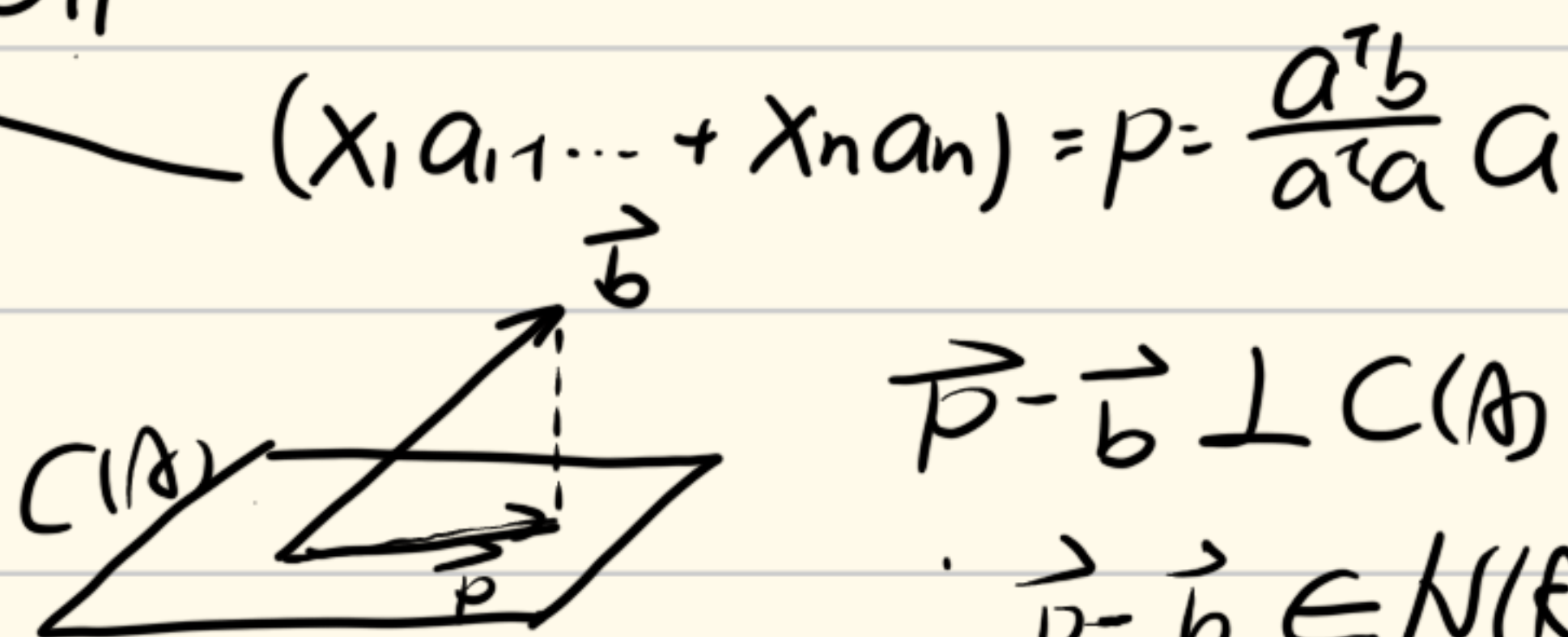
余弦定理: $\cos \theta = \frac{x^T y}{\|x\| \|y\|}$

投影矩阵: $P = \frac{a a^T}{a^T a}$

投影后向量 $p = \frac{a^T b}{a^T a} a$

最小二乘

$$E^2 = \|Ax - b\|^2$$



$$(x_1 a_1 + \dots + x_n a_n) = p = \frac{a^T b}{a^T a} a$$

$$\vec{p} - \vec{b} \perp C(A)$$

$$\therefore \vec{p} - \vec{b} \in N(A^T)$$

$$A^T A x = A^T b$$

$$A^T (\vec{p} - \vec{b}) = A^T A x - A^T b = 0$$

$$1. a \perp b \quad a^T b = 0$$

$$\downarrow$$

$$A \perp B, \forall a \in A, b \in B, a \perp b$$

$$A \subseteq V \quad \{v \mid v \in V, v \perp A\} \equiv A^\perp$$

$$C(A^T) = N(A)^\perp \subseteq \mathbb{R}^n$$

$$C(A) = N(A^T)^\perp \subseteq \mathbb{R}^m$$

$$2. b \text{ 在 } a \text{ 上的投影} \quad p = \frac{a^T b}{a^T a} a$$

$$P = \frac{aa^T}{a^T a} \quad Pb = p$$

$$3. Ax = b$$

$$\text{单变量: } ax = b \quad \|ax - b\|^2$$

$$ax = a \frac{a^T b}{a^T a} \quad x = \frac{a^T b}{a^T a}$$

$$\text{多变量: } A^T A x = A^T b$$

$$P = A(A^T A)^{-1} A^T$$

$$\text{带权重: } A^T W^T W A x = A^T W^T W b$$

规范正交/正交归一 orthonormal

$$\{q_i\}, \quad q_i^T q_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

orthonormal basis

正交矩阵: 方阵 Q $Q^T Q = Q Q^T = I$ ($Q^T = Q^{-1}$)

$$\textcircled{1} Q = [q_1 \dots q_n] \quad Q^T Q = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} [q_1 \dots q_n] = M = [m_{ij} = q_i^T q_j]_{n \times n}$$

$\{q_i\}$ 是 $C(Q)$ 的 orthogonal basis $\Rightarrow q_i^T q_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

$$r = n = m \quad \text{span}\{q_i\} = C(Q) = \mathbb{R}^n$$

$$\textcircled{2} \|Qx\| = \|x\| \quad \text{保长度} \quad \langle Qx, Qx \rangle = (Qx)^T Qx \\ = x^T Q^T Q x = x^T x$$

$$\textcircled{3} (Qx)^T Qy = x^T y \quad \text{保内积}$$

orthonormal basis:

$$\textcircled{1} b = x_1 q_1 + x_2 q_2 + \dots + x_n q_n$$

$$q_i^T b = x_i q_i^T q_i = x_i$$

$$b = (q_1^T b) q_1 + (q_2^T b) q_2 + \dots + (q_n^T b) q_n$$

$$\textcircled{2} p_i = \frac{q_i q_i^T}{q_i^T q_i} \quad p_i = \frac{q_i q_i^T b}{q_i^T q_i} = q_i (q_i^T b)$$

$$b = p_1 + p_2 + \dots + p_n$$

Gram-Schmidt orthonormalization:

$\{v_i\} \rightarrow$ 正交归一基

$$q_i = \frac{v_i}{\|v_i\|}$$

$$q_{i+1} = \frac{v_{i+1} - p_{i+1}}{\|v_{i+1} - p_{i+1}\|}$$

$$p_{i+1} = (q_1^T v_{i+1}) q_1 + \dots + (q_i^T v_{i+1}) q_i$$

QR factorization: 对任意满秩矩阵
列

$$Q = [q_1 \dots q_n]$$

$$R = \begin{bmatrix} \|v_1\| & q_1^T v_2 \\ & \|v_2 - (q_1^T v_2) q_1\| \end{bmatrix}$$

$$q_2 = \frac{v_2 - (q_1^T v_2) q_1}{\|v_2 - (q_1^T v_2) q_1\|}$$

↓

$$v_2 = \|v_2 - (q_1^T v_2) q_1\| q_2 + (q_1^T v_2) q_1$$

$$A^T A x = A^T b$$

$$A = QR \quad R^T Q^T Q R x = R^T Q^T b$$

$$R^T R x = R^T Q^T b \quad R \text{ 可逆}, R^T \text{ 可逆}$$

$$R x = Q^T b \quad \text{回代} \checkmark$$

bilinear form: $V \times V \rightarrow \mathbb{R}$
 $(v, w) \rightarrow (v, w)$

两个位置都是线性的: $\langle v, \alpha w_1 + \beta w_2 \rangle = \alpha \langle v, w_1 \rangle + \beta \langle v, w_2 \rangle$

$$\langle \alpha v_1 + \beta v_2, w \rangle = \alpha \langle v_1, w \rangle + \beta \langle v_2, w \rangle$$

Symmetric: $\langle v, w \rangle = \langle w, v \rangle$ ($\mathbb{R}$ 范围内)

inner product: 满足 $\langle v, v \rangle \geq 0$ 的对称 bilinear form

行列式 determinant:

定义

- ① $\det(1) = 1$
- ② 若两行相等 行列式为0
- ③ $A = \begin{bmatrix} q_1' v_1' + q_1'' v_1'' \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad |A| = q_1' \begin{bmatrix} v_1' \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + q_1'' \begin{bmatrix} v_1'' \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

Type I

④ 交换行 $\Rightarrow$ 变符号

⑤ Type II 不变行列式

⑥ 如果有一行为0 则行列式为0

⑦ $\det(A) \neq 0$ A invertible/nonsingular

⑧ $\det(A) = 0$ A singular

⑨ $\det(A) = \det(A^T)$

⑩ $\det(AB) = \det(A) \det(B)$

显式表达式: $\det(A) = \sum_{P=(i_1, i_2, \dots, i_n)} a_{1i_1} \dots a_{ni_n} \det(P) = (-1)^{\text{inv}(P)}$

Cofactor 代数余子式:

$$C_{ij} = (-1)^{i+j} \det([A]_{ij})$$

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$$

在第1行展开

$$\det(A) = a_{j1}C_{j1} + a_{j2}C_{j2} + \dots + a_{jn}C_{jn}$$

在第j列展开

$$A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad x_j = \frac{\det(B_j)}{\det(A)} \quad B_j: \text{将 } A \text{ 的第 } j \text{ 列替换为 } B$$

pivot 可以用A的子矩阵行列式表示:

$$d_k = \frac{|A_k|}{|A_{k-1}|}$$

Type III 处理后的 pivot

$$|A + \alpha\beta^T| = |A|(1 + \beta^T A^{-1} \alpha) \quad \text{秩-1矩阵的扰动}$$

$$A^T A = R^T Q^T Q R = R^T R \quad R = DU$$

$$= U^T D^2 U$$

$$A^T A U^{-1} = U^T D^2$$

$$A = QR = QDU$$

$$AU^{-1} = QD$$

elimination on $A^T A$

column 1: $\|d_1\|$

column 2: $\|d_2\|$
.....

$$\begin{cases} B = a_2 + \frac{1}{2}a_1 \\ C = a_3 + \frac{2}{3}a_2 + \frac{1}{3}a_1 \end{cases}$$

$$A^T A = R^T \underbrace{Q^T Q}_I R = R^T R$$

$$= \underbrace{U^T D^2 U}$$

$$A^T A U^{-1} = \underbrace{U^T D^2}$$

$$A = QR = QDU$$

$$\underline{AU^{-1} = QD}$$

$R = DU$
 对角 主元是1的上三角

pivot: $\begin{bmatrix} d_{11}^2 & & \\ * & d_{22}^2 & \\ * & * & d_{33}^2 \end{bmatrix}$

$$\begin{bmatrix} 0 & 1 & \dots & 1 \\ 1 & 0 & & \vdots \\ & & 0 & \vdots \\ & & & \ddots & 1 \\ 1 & \dots & & & 0 \end{bmatrix}$$

行列式

$$1. \sum_{P=(i_1, i_2, \dots, i_n)} a_{1i_1} a_{2i_2} \dots a_{ni_n} \det(P) = (-1)^{\text{inv}(P)}$$

$$2. \text{cofactor: } C_{ij} = (-1)^{i+j} \det([-i])$$

$$\det(A) = a_{1j} C_{1j} + \dots + a_{nj} C_{nj}$$

$$= a_{11} C_{11} + \dots + a_{in} C_{in}$$

$$3. A^{-1} = \frac{C^T}{\det(A)} \rightarrow [C_{ij}^T = C_{ji}]_{n \times n}$$

$$4. Ax=b \quad X = \frac{1}{\det(A)} \begin{bmatrix} \det(A(a_1 \rightarrow b)) \\ \det(A(a_2 \rightarrow b)) \\ \vdots \\ \det(A(a_n \rightarrow b)) \end{bmatrix}$$

5. pivot 公式: A invertible, 仅 Type III 变为 U

$$d_i = \frac{\det(A_i)}{\det(A_{i-1})}$$

$$A = \begin{bmatrix} Q_1 & * & \dots & * \\ & Q_2 & & \vdots \\ & & \ddots & x \\ & & & Q_n \end{bmatrix} \quad \det(A) = \det(Q_1) \det(Q_2) \dots \det(Q_n)$$

(分块化 upper triangular 计算, 不能用 $AD-BC$)

$$6. \text{若 } A \text{ 可逆} \quad |A + \alpha \beta^T| = |A| (1 + \beta^T A^{-1} \alpha)$$

$$\begin{vmatrix} 1 & 1 & \dots & 1 \\ x_0 & x_1 & \dots & x_{n-1} \\ x_0^2 & x_1^2 & \dots & x_{n-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_0^{n-1} & \dots & \dots & x_{n-1}^{n-1} \end{vmatrix} = \prod_{n-1 \geq j > i \geq 1} (x_j - x_i)$$

特征值与特征向量:

$$1. \text{求特征值: } (A - \lambda_i I)X = 0$$

$$|A - \lambda_i I| = 0 \Rightarrow \lambda_1, \lambda_2, \dots, \lambda_k$$

$$2. \text{求特征向量: } (A - \lambda_i I)X = 0$$

$$A[x_1 \dots x_n] = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} [x_1 \dots x_n]$$

特征值, 向量

$$Ax = \lambda x \quad (x \neq 0) \quad (A - \lambda I)x = 0 \quad x \neq 0$$

$$\textcircled{1} |A - \lambda I| = 0 \Rightarrow \lambda_1, \lambda_2, \dots, \lambda_n$$

$$\textcircled{2} (A - \lambda_1 I)x = 0 \Rightarrow x_i = C_1 v_1 + C_2 v_2 + \dots + C_n v_n \quad (C_1, C_2, \dots, C_n \neq 0)$$

$$\textcircled{3} \text{迹 trace (方阵)} \quad \text{tr}(A) = \sum_{i=1}^n a_{ii}$$

$$\textcircled{4} \text{tr}(A) = \sum_{i=1}^n \lambda_i \quad |A| = \prod_{i=1}^n \lambda_i$$

对角化:

$$\textcircled{1} \text{若 } A \text{ 有 } n \text{ 个线性无关 eigenvector } \{v_1, \dots, v_n\} \quad S = [v_1 \dots v_n]$$

$$S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

$$\textcircled{2} \text{所有 } \lambda_i \text{ distinct} \Rightarrow A \text{ diagonalizable (充分不必要)}$$

$$\textcircled{3} Av_i = \lambda_i v_i \Rightarrow f(A)v_i = f(\lambda_i)v_i$$

$$\textcircled{4} A \text{ 与 } B \text{ 有相同的 } S \Leftrightarrow AB = BA$$

矩阵对角化的应用:

迭代过程:

$$U_0 = S \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$\begin{cases} U_{k+1} = AU_k \\ U_0 \end{cases}$$

$$U_k = A^k U_0 = c_1 \lambda_1^k X_1 + \dots + c_k \lambda_k^k X_k$$

$$A = S \Lambda S^{-1}$$

$$S = [v_1 \ v_2 \ \dots \ v_n] \quad U_0 = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

$$U_k = S \Lambda^k S^{-1} U_0 = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$= \sum_{i=1}^k c_i \lambda_i^k X_i$$

$$U_n = AU_{n-1}$$

① stable $\forall |\lambda_i| < 1$

② neutrally stable some $|\lambda_i| = 1$, others < 1

③ unstable $\exists \lambda_i, |\lambda_i| > 1$

复数:

1. 复向量空间 $C^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mid x_i \in \mathbb{C} \right\}$

$$\bar{x}(x^*) = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_n \end{bmatrix} \quad \bar{x}^T x \equiv \|x\|^2, \quad \langle x, y \rangle = \bar{x}^T y$$

$$\text{正交 } \langle x, y \rangle = \bar{x}^T y = 0$$

2. 复矩阵 $A = (a_{ij})_{m \times n} \quad a_{ij} \in \mathbb{C}$

$$\bar{A} = (\bar{a}_{ij}) \quad \overline{AB} = \bar{A} \bar{B} \quad \overline{\lambda A} = \bar{\lambda} \bar{A}$$

$$\text{共轭转置} \quad A^H = (\bar{A})^T = \bar{A}^T$$

(厄米矩阵)

$$(AB)^H = B^H A^H$$

3. Hermitian matrix $A = A^H$

$$\textcircled{1} x^H A x \in \mathbb{R} \quad (x^H A x)^H = x^H A x$$

$$\textcircled{2} Ax = \lambda x, \quad \lambda \in \mathbb{R}$$

$$\textcircled{3} Ax_1 = \lambda_1 x_1 \quad Ax_2 = \lambda_2 x_2$$

$$\lambda_1 \neq \lambda_2 \Rightarrow x_1^H x_2 = 0$$

$$x_1^H A x_2 \Rightarrow \lambda_2 x_1^H x_2 = \lambda_1 x_1^H x_2$$

4. 酉矩阵 $U^H U = U U^H = I$

$$\textcircled{1} (Ux)^H (Uy) = x^H U^H U y = x^H y \quad (\text{保长度, 保内积})$$

$$\textcircled{2} Ux = \lambda x \Rightarrow |\lambda| = 1 \quad \lambda = \pm 1$$

$$\textcircled{3} Ux_1 = \lambda_1 x_1 \quad Ux_2 = \lambda_2 x_2 \quad \lambda_1 \neq \lambda_2 \Rightarrow x_1^H x_2 = 0 (\perp)$$

5. skew-Hermitian matrix $K^H = -K$

① if $A = A^H$, $(iA)^H = iA$ (iA Hermitian)

② eigenvalues are pure imaginary

③ $KX_1 = \lambda_1 X_1$ $KX_2 = \lambda_2 X_2$ $\lambda_1 \neq \lambda_2 \Rightarrow X_1^H X_2 = 0$

相似矩阵:

1. $A \sim B$ $B = M^{-1}AM$

2. 若 $B = M^{-1}AM$, 则 A 与 B 有相同 eigen value

且 if $AX = \lambda X$, each eigenvector X of A ,
 $BM^{-1}X$ $M^{-1}X$ is a eigenvector of B .

几何意义: $[V_1 \ V_2 \ \dots \ V_n] = [W_1 \ W_2 \ \dots \ W_n]M$

$$Q([V_1 \ \dots \ V_n]) = [V_1 \ \dots \ V_n]A$$

$$U^H A U$$

4. $\forall A$, $\exists U$ (unitary), $U^{-1}AU$ upper triangular

5. 若 $A = A^H$, $\exists U$, $U^{-1}AU = U^H A U = \Lambda$ (real diagonal)

① 若 A 实对称, U 为正交矩阵 Q , $A = Q\Lambda Q^{-1}$

$$A^H = A \Rightarrow U^H A U = D \quad (U^H = U^{-1})$$

$\downarrow$ unitary $\downarrow$ diagonal
 $(Q^T = Q^{-1})$

real: $A^T = A \Rightarrow Q^T A Q = D$

正规矩阵: $A^H A = A A^H$ $\downarrow$ 正规矩阵

$$\Leftrightarrow \exists U, U^H A U = D$$

Jordan form:

Jordan block: $J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}$

Jordan form: $J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_i \end{bmatrix}$

定理: 若矩阵 A 有 S 个线性无关的本征向量

$$A \sim J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_s \end{bmatrix}$$

$$ax^2 + 2bxy + cy^2 > 0$$

$$a = \frac{\partial^2 F}{\partial x^2} \quad b = \frac{\partial^2 F}{\partial x \partial y} \quad c = \frac{\partial^2 F}{\partial y^2}$$

$a > 0, ac - b^2 > 0$ positive definite

$$[x_1 \ x_2 \ \dots \ x_n] A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \sum_{i,j} a_{ij} x_i x_j$$

正定充要条件:

1. 对称矩阵 A : $\forall x \neq 0, x^T A x > 0$

2. $\forall \lambda_i > 0, A = \lambda_1 v_1 v_1^T + \dots + \lambda_n v_n v_n^T$

3. $\forall |A_k| > 0$

4. $\forall d_k > 0$ (无行交换)

5. $\exists \text{ inv } R \text{ s.t. } A = R^T R$ $\left\{ \begin{array}{l} A = LDL^T = L\sqrt{D}\sqrt{D}^T L^T \\ A = Q\Lambda Q^T = Q\sqrt{\Lambda}\sqrt{\Lambda}^T Q^T \\ A = Q\sqrt{\Lambda}Q^T Q\sqrt{\Lambda}Q^T \end{array} \right.$
 $A = R^T R = R^T Q^T Q R = (QR)^T QR$

半正定:

1. $A \geq 0 \quad \forall x \neq 0, x^T A x \geq 0$

2. $\lambda_i \geq 0$

3. $|A_I| \geq 0, 1 \leq \{1 < 2 < \dots < n\}$

4. $d_i \geq 0$

5. $\exists R$ (不一定可逆), $A = R^T R$

惯性定律:

1. 合同变换 (congruence trans) $A \rightarrow C^T A C$

2. $C^T A C$ 与 A 有相同 $>0 / =0 / <0$ 的 λ 个数

$$A = S^{-1} D S \quad D = S A S^{-1}$$

$$r(A) = q \Rightarrow \dim N(A) = n - q$$

3. 二次超曲面的判定

$$x^T A x = 1$$

$$x^T Q^T \Lambda Q x = 1$$

$$y = Qx \Rightarrow y^T \Lambda y = 1$$

A 对称实矩阵 $\Rightarrow \exists$ 正交矩阵 Q , $A = Q \Lambda Q^T$

保长度, 保角度

1.7, 2.5, 3.5, 5.4, 6.4, 6.5 不考

Chapter 3

① Orth: $x^T y = 0 \Rightarrow$ 子空间正交 $\begin{cases} C(A) \perp N(A^T) \\ C(A^T) \perp N(A) \end{cases}$
 $x^H y = 0$

② Projection $\begin{cases} ① P^2 = P \\ ② P^T = P \end{cases}$ $V: P = \frac{V V^T}{V^T V} \Rightarrow W, P = P_W = \frac{V V^T}{V^T V} W = \frac{V^T W}{V^T V} V = \frac{V^T}{V^T V} W \frac{V}{\sqrt{V^T V}}$
 $C(A): P = A(A^T A)^{-1} A^T$

$$Ax = b \quad \|b - Ax\|^2 \text{ minimized}$$

$$x = A(A^T A)^{-1} A^T b$$

③ Orthonormal basis

Schmit $q_1 = \frac{v_1}{\|v_1\|} \quad q_2 = \frac{v_2 - (q_1^T v_2) q_1}{\|v_2 - (q_1^T v_2) q_1\|} \quad \dots$

$$[v_1 \dots v_n] = [q_1 \dots q_n] \begin{bmatrix} \|v_1\| & q_1^T v_2 & \dots & q_1^T v_n \\ & \|x_2\| & & \vdots \\ & & \ddots & \vdots \\ & & & \|x_n\| \end{bmatrix} \quad QR \text{ decomp}$$

Chapter 4

行列式性质:

$$A \text{ 可逆} \Leftrightarrow |A| \neq 0$$

$$A \sim U \quad |A| = |U| \quad (\text{仅 Type III})$$

$$|U| = \prod_{i=1}^n d_i$$

$$\det(AB) = \det(A) \det(B)$$

② Cofactor: $C_{ij} = |A(\text{去 } i \text{ 行 } j \text{ 列})| \times (-1)^{i+j}$

$$|A| = a_{i1} C_{i1} + \dots + a_{in} C_{in}$$

$$= a_{1j} C_{1j} + \dots + a_{nj} C_{nj}$$

$$\textcircled{3} A^{-1} = \frac{1}{|A|} C^T \quad C = [C_{ij}]_{n \times n}, C^T = [C^T_{ij} = C_{ji}]_{n \times n}$$

$$\Rightarrow \text{Cramer's rule } X = A^{-1}b = \frac{1}{|A|} \begin{bmatrix} B_1 \\ \vdots \\ B_n \end{bmatrix}, B_i = |A \text{ (第 } i \text{ 列替换成 } b)|$$

$$\textcircled{4} d_k = \frac{|A_k|}{|A_{k-1}|}$$

Chapter 5

$$Ax = \lambda x \quad (x \neq 0)$$

$$\textcircled{1} |A - \lambda I| = 0 \Rightarrow \lambda_1, \dots, \lambda_n$$

$$\textcircled{2} \lambda_i: Ax = \lambda_i x \quad x_i$$

$\textcircled{3}$ 如果有 n 个线性无关 eigenvector.

$$S = [v_1 \dots v_n]$$

$$AS = [Av_1 \dots Av_n] = [v_1 \dots v_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

$$\Rightarrow S^{-1}AS = \Lambda \Rightarrow A = SAS^{-1}$$

$$A^k = (S\Lambda S^{-1})^k = S\Lambda^k S^{-1}$$

$$\begin{cases} a_{n+2} = q_1 a_{n+1} + q_2 a_n \\ a_{n+1} = a_{n+1} \end{cases} \quad \begin{bmatrix} a_{n+2} \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} q_1 & q_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_{n+1} \\ a_n \end{bmatrix}$$

$$V_{n+1} = A V_n$$

$$V_n = A^n V_0$$

$$\textcircled{4} A^H = (\bar{A})^T = \bar{A}^T \quad (AB)^H = B^H A^H$$

$$\text{Hermitian: } A^H = A \quad \text{Unitary: } U^H = U^{-1} / U^H U = U U^H = I$$

$\textcircled{5}$ Similar transf

$$A \rightarrow M^{-1} A M = B$$

$$B \sim A \quad \begin{matrix} \text{eigenvalues 相同} \\ Ax_i = \lambda_i x_i \Rightarrow B M^{-1} x_i = \lambda_i (M^{-1} x_i) \end{matrix}$$

⑥ Spectral thm $A^H = A : \exists U \text{ s.t. } A = U \Lambda U^H$ ^{diagonal}

real $A^T = A : \exists Q \text{ s.t. } A = Q \Lambda Q^T$

$AA^H = A^H A \quad \exists U \text{ s.t. } A = U \Lambda U^H$

⑦ Jordan form:

(1) Jordan block $J^k = \begin{bmatrix} \lambda_1 & 1 & & 0 \\ & \ddots & \ddots & \\ 0 & & \lambda_k & 1 \end{bmatrix}$

Thm: $\forall A$ 有 n 个线性无关 eigenvectors

$\exists \text{ inv } R \text{ s.t. } R^{-1} A R = \begin{bmatrix} J^1 & & \\ & \ddots & \\ & & J^k \end{bmatrix}$

求法: A ①求 A 的 eigenvalues 与 eigenvectors

$\Rightarrow$ 最大线性无关 eigenvector 数量